

POINTER PROGRAMS

Questions

1. Calculate average of array using pointers.
2. Count vowels in string using pointers.
3. Find minimum element in array using pointers.
4. Reverse array using pointers.
5. Perform matrix addition using pointers.

Answers

1. Average of Array

```
#include <stdio.h>
int main() {
    int arr[5] = {10,20,30,40,50};
    int *p = arr;
    int sum = 0;

    for(int i=0;i<5;i++) {
        sum += *(p+i);
    }

    printf("Average = %.2f", (float)sum/5);
    return 0;
}
```

2. Count Vowels

```
#include <stdio.h>
int main() {
    char str[] = "Hello World";
    char *p = str;
    int count = 0;

    while(*p != '\0') {
        if(strchr("aeiouAEIOU", *p)) count++;
        p++;
    }

    printf("Vowels = %d", count);
    return 0;
}
```

3. Minimum Element

```
#include <stdio.h>
int main() {
    int arr[5] = {12,5,8,20,3};
    int *p = arr;
    int min = *p;
```

```

    for(int i=1;i<5;i++) {
        if(*(p+i) < min)
            min = *(p+i);
    }

    printf("Minimum = %d", min);
    return 0;
}

```

4. Reverse Array

```

#include <stdio.h>
int main() {
    int arr[5] = {1,2,3,4,5};
    int *start = arr;
    int *end = arr+4, temp;

    while(start < end) {
        temp = *start;
        *start = *end;
        *end = temp;
        start++;
        end--;
    }

    for(int i=0;i<5;i++)
        printf("%d ", arr[i]);

    return 0;
}

```

5. Matrix Addition

```

#include <stdio.h>
int main() {
    int a[2][2]={1,2},{3,4}, b[2][2]={5,6},{7,8}, c[2][2];
    int *p1=&a[0][0], *p2=&b[0][0], *p3=&c[0][0];

    for(int i=0;i<4;i++)
        *(p3+i)=*(p1+i)+*(p2+i);

    for(int i=0;i<2;i++) {
        for(int j=0;j<2;j++)
            printf("%d ", c[i][j]);
        printf("\n");
    }
    return 0;
}

```